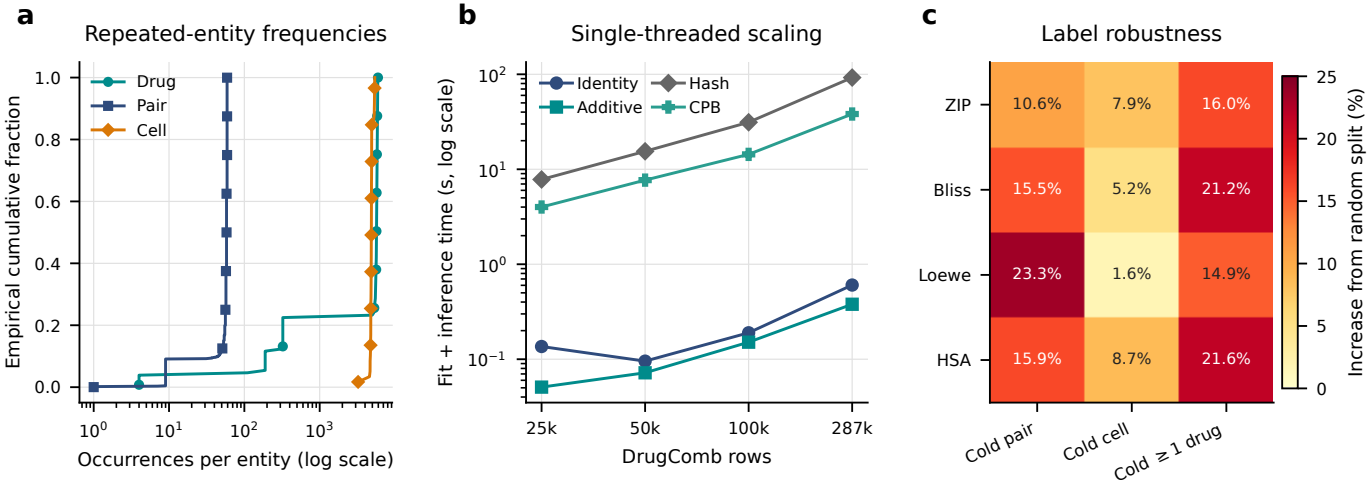
a
Repeated-entity frequencies
Drug
Pair
Cell
Empirical cumulative fraction
Occurrences per entity (log scale)
b
Single-threaded scaling
Identity
Hash
Additive
CPB
Fit + inference time (s, log scale)
DrugComb rows
25k
50k
100k
287k
c
Label robustness
ZIP
Bliss
Loewe
HSA
10.6%
7.9%
16.0%
15.5%
5.2%
21.2%
23.3%
1.6%
14.9%
15.9%
8.7%
21.6%
Cold pair
Cold cell
Cold ≥ 1 drug
Increase from random split (%)